

This section provides knowledge and understanding of the atom, nucleus, fundamental particles, radioactivity, fission and fusion.

The building of nuclear power stations can be used to evaluate the benefits and risks to society (HSW9). Ethical, environmental and decision making issues may also be discussed (HSW10 and HSW12).

6.4.1 The nuclear atom

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) alpha-particle scattering experiment; evidence of a small charged nucleus	
(b) simple nuclear model of the atom; protons, neutrons and electrons	
(c) relative sizes of atom and nucleus	M0.4, M1.4
(d) proton number; nucleon number; isotopes; notation A_ZX for the representation of nuclei	
(e) strong nuclear force; short-range nature of the force; attractive to about 3 fm and repulsive below about 0.5 fm	1 fm = 10^{-15} m
(f) radius of nuclei; $R = r_0A^{1/3}$ where r_0 is a constant and A is the nucleon number	
(g) mean densities of atoms and nuclei.	